School of Computer Science Engineering & Information Systems

Department of Software & Systems Engineering

ISWE103P – Database Systems

CYCLE SHEET– I

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LAB SLOT:L37+L38

I. Create the following tables Company Database
Employee (SSN, Name (fname, Minit, Lname), Sex, Address, salary, DOB, Department, designation, SupervisorSSN)

```
Enter user-name: SYSTEM
Enter password:

Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

SQL> create table employee(SSN varchar(10),fname varchar(10),minit varchar(10),lname varchar(10),sex varchar(3),address varchar(15),salary number(15,5),dob
date,department varchar(15),designation varchar(15),SUPERVISORSSN varchar(10));

Table created.
```

```
SQL> desc employee;
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------------------
----
 SSN                                                VARCHAR2(10)
 FNAME                                              VARCHAR2(10)
 MINIT                                              VARCHAR2(10)
 LNAME                                              VARCHAR2(10)
 SEX                                                VARCHAR2(3)
 ADDRESS                                            VARCHAR2(15)
 SALARY                                             NUMBER(15,5)
 DOB                                                DATE
 DEPARTMENT                                         VARCHAR2(15)
 DESIGNATION                                        VARCHAR2(15)
 SUPERVISORSSN                                      VARCHAR2(10)

SQL>
```

Department (Dep_num, Name, ManagerSSN, Manager_DOB, Location)

```
SQL> create table department_new(department_num number(10),depname varchar(20),managerssn varchar(10),man_dob date,location varchar(20));

Table created.
```

```
SQL> desc department_new;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DEPARTMENT_NUM                                     NUMBER(10)
 DEPNAME                                            VARCHAR2(20)
 MANAGERSSN                                         VARCHAR2(10)
 MAN_DOB                                            DATE
 LOCATION                                           VARCHAR2(20)
```

Dependent (Name, DOB, Sex, Relationship, Employee SSN)

```
SQL> create table dependent(name varchar(20),dob date,sex varchar(2),relationship varchar(10),employeessn varchar(10));

Table created.

SQL>

SQL> desc dependent;
 Name                                      Null?    Type
 ----------------------------------------- -------- ---------------------------
 NAME                                               VARCHAR2(20)
 DOB                                                DATE
 SEX                                                VARCHAR2(2)
 RELATIONSHIP                                       VARCHAR2(10)
 EMPLOYEESSN                                        VARCHAR2(10)
```

Project (Proj_num, Name, Location, Controlling Department, Budget)

```
SQL> create table project(project_num number(10),name varchar(20),location varchar(15),controlling_department varchar(20),budget number(15,3));

Table created.
```

```
SQL> desc project;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PROJECT_NUM                                        NUMBER(10)
 NAME                                               VARCHAR2(20)
 LOCATION                                           VARCHAR2(15)
 CONTROLLING_DEPARTMENT                             VARCHAR2(20)
 BUDGET                                             NUMBER(15,3)
```

Works_on (SSN, Project Num, Hours)

```
SQL> create table works_on(ssn varchar(10),project_num number(10),hours number(5));

Table created.

SQL> desc works_on;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SSN                                                VARCHAR2(10)
 PROJECT_NUM                                        NUMBER(10)
 HOURS                                              NUMBER(5)
```

a) Identify the appropriate primary keys for the tables.

```
SQL> alter table employee add constraint pk_employee primary key(ssn);

Table altered.

SQL> desc employee;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SSN                                       NOT NULL VARCHAR2(10)
 FNAME                                              VARCHAR2(10)
 MINIT                                              VARCHAR2(10)
 LNAME                                              VARCHAR2(10)
 SEX                                                VARCHAR2(3)
 ADDRESS                                            VARCHAR2(15)
 SALARY                                             NUMBER(15,5)
 DOB                                                DATE
 DEPARTMENT                                         VARCHAR2(15)
 DESIGNATION                                        VARCHAR2(15)
 SUPERVISORSSN                                      VARCHAR2(10)
```

```
SQL> alter table department_new add constraint pk_department_new primary key
(department_num);

Table altered.

SQL> desc department_new;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DEPARTMENT_NUM                            NOT NULL NUMBER(10)
 DEPNAME                                            VARCHAR2(20)
 MANAGERSSN                                         VARCHAR2(10)
 MAN_DOB                                            DATE
 LOCATION                                           VARCHAR2(20)
```

```
SQL> alter table dependent add constraint pk_dependent primary key(employees
sn);

Table altered.

SQL>
SQL> desc dependent;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 NAME                                               VARCHAR2(20)
 DOB                                                DATE
 SEX                                                VARCHAR2(2)
 RELATIONSHIP                                       VARCHAR2(10)
 EMPLOYEESSN                               NOT NULL VARCHAR2(10)
```

```
SQL> alter table project add constraint pk_project primary key(project_num);


Table altered.

SQL> desc project;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PROJECT_NUM                               NOT NULL NUMBER(10)
 NAME                                               VARCHAR2(20)
 LOCATION                                           VARCHAR2(15)
 CONTROLLING_DEPARTMENT                             VARCHAR2(20)
 BUDGET                                             NUMBER(15,3)
```

```
SQL> alter table works_on add constraint pk_works_on primary key(ssn,project
_num);

Table altered.

SQL> desc works_on;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SSN                                       NOT NULL VARCHAR2(10)
 PROJECT_NUM                               NOT NULL NUMBER(10)
 HOURS                                              NUMBER(5)
```

b) Insert at least 10 rows to each table.

```
SQL> select * from employee;

SSN        FNAME      MINIT      LNAME      SEX ADDRESS            SALARY DOB       DEPARTMENT      DESIGNATION     SUPERVISOR
---------- ---------- ---------- ---------- --- --------------- ---------- --------- --------------- --------------- ----------
a1         sabari     j          eswar      m   cuddalore           50000 01-MAY-90 software        architect       a1
a2         sanjay     s          saravanan  m   salem               45000 12-JUN-89 ai              analyst         s2
a3         sanjika    s          senthil    f   krishnagiri         90000 24-MAY-89 hr              manager         s3
a4         adi        g          govind     m   chennai             67000 10-DEC-87 marketing       assistant       s4
a5         harish     v          velan      m   karur               71000 11-NOV-92 it              developer       s5
a6         prabha     s          sugan      f   vellore             95000 01-JAN-85 cybersecurity   hacker          s6
a7         akshay     s          sugadevan  m   krishnagiri         80000 04-DEC-93 it              analyst         s7
a8         barath     d          devan      m   katpadi             60000 21-JUN-85 software        developer       s8
a9         arya       d          kumar      m   ariyalur            67000 19-NOV-87 finance         assistant       s9
a10        mowni      m          madhavan   f   dharmapuri          75000 06-MAR-81 hr              manager         s10
```

```
SQL> select * from department_new;

DEPARTMENT_NUM DEPNAME              MANAGERSSN MAN_DOB   LOCATION
-------------- -------------------- ---------- --------- --------------------
             1 production           m1         04-FEB-87 london
             2 it                   m2         29-MAR-92 new york
             3 cybersecurity        m3         10-APR-91 syndey
             4 hr                   m4         17-JAN-87 mumbai
             5 sales                m5         30-NOV-78 delhi
             6 production           m6         26-JAN-94 barcelona
             7 software             m7         15-AUG-81 chennai
             8 ai                   m8         08-SEP-89 chicago
             9 finance              m9         14-OCT-90 dubai
            10 marketing            m10        18-NOV-84 paris

10 rows selected.
```

```
SQL> select  * from dependent;

NAME                 DOB       SE RELATIONSH EMPLOYEESS
-------------------- --------- -- ---------- ----------
ezhil                01-DEC-89 f  friend     d1
pavan                02-JAN-92 m  friend     d2
sugan                18-JUN-52 m  father     d3
prabha               27-APR-58 f  mother     d4
tamil                06-DEC-16 m  son        d5
hari                 02-FEB-40 m  uncle      d6
govind               25-MAR-93 m  friend     d7
samji                12-OCT-18 f  daughter   d8
dev                  13-AUG-17 m  son        d9
lassika              21-APR-57 f  parent     d10
```

```
SQL> select * from project;

PROJECT_NUM NAME                 LOCATION       CONTROLLING_DEPARTME     BUDGET
----------- -------------------- -------------- -------------------- ----------
          1 rocket               chennai        101                      200000
          2 dc21                 bangalore      102                      500000
          3 lr023                kochi          103                      340000
          4 kishi2               nagaland       104                      780000
          5 lass82               assam          105                      640000
          6 bharat               bihar          106                      840000
          7 saqu9                mumbai         107                      193300
          8 supra                kashmir        108                      173000
          9 airindia             kolkata        109                      647000
         10 misew2               delhi          110                      353000

10 rows selected.
```

c) Set the following Constraints to the above tables:

1. Department number should be in the range 1000 to 2000.

```
SQL> select * from works_on;

SSN        PROJECT_NUM      HOURS
---------- ----------- ----------
001                  1         29
002                  2         65
3                    3         56
004                  4         18
005                  5         45
006                  6         34
007                  7         43
008                  8         20
009                  9         52
010                 10         30

10 rows selected.
```

c) Set the following Constraints to the above tables:

1. Department number should be in the range 1000 to 2000.

```
SQL> ALTER TABLE DEPARTMENT
  2  ADD CONSTRAINT chk_department_number
  3  CHECK (DEP_NUM BETWEEN 1000 AND 2000);

Table altered.
```

2.Relationship of the dependents to an employee should be only Spouse, Son, Daughter, and Parent.

```
SQL> ALTER TABLE DEPENDENT
  2  ADD CONSTRAINT chk_relationship
  3  CHECK (RELATIONSHIP IN ('Spouse', 'Son', 'Daughter', 'Parent'));

Table altered.
```

3. Name of the project doesn't exceed 4 characters.

```
SQL> ALTER TABLE PROJECT
  2  ADD CONSTRAINT chk_project_name
  3  CHECK (LENGTH(NAME) <= 4);

Table altered.
```

4. Employee salary should be above 1000.

```
SQL> alter table employee
  2  add constraint chk_employee_salary
  3  check (salary>1000);

Table altered.
```

5. The location of the project is mandatory.

```
SQL> alter table project
  2  modify location varchar(50) not null;

Table altered.

SQL> |
```

6. The minimum hour of a project is 20 hours and maximum 120 hours.

```
SQL> ALTER TABLE Works_on
  2  ADD CONSTRAINT chk_hours
  3  CHECK (Hours BETWEEN 20 AND 120);

Table altered.
```

7. The department names can be 'IT', 'HR', 'Sales', 'Manufacturing' and 'Production'.

```
SQL> alter table department add constraint chk_department_name check(name in
('it','hr','sales','manufacturing','production'));

Table altered.

SQL>
```

d) Query the Database

8. Display all the employee details of the employee table

```
SQL> select * from employee;

SSN        FNAME      MINIT      LNAME      SEX ADDRESS            SALARY DOB       DEPARTMENT      DESIGNATION     SUPERVISOR
---------- ---------- ---------- ---------- --- --------------- ---------- --------- --------------- --------------- ----------
a1         sabari     j          eswar      m   cuddalore           50000 01-MAY-90 software        architect       a1
a2         sanjay     s          saravanan  m   salem               45000 12-JUN-89 ai              analyst         s2
a3         sanjika    s          senthil    f   krishnagiri         90000 24-MAY-89 hr              manager         s3
a4         adi        g          govind     m   chennai             67000 10-DEC-87 marketing       assistant       s4
a5         harish     v          velan      m   karur               71000 11-NOV-92 it              developer       s5
a6         prabha     s          sugan      f   vellore             95000 01-JAN-85 cybersecurity   hacker          s6
a7         akshay     s          sugadevan  m   krishnagiri         80000 04-DEC-93 it              analyst         s7
a8         barath     d          devan      m   katpadi             60000 21-JUN-85 software        developer       s8
a9         arya       d          kumar      m   ariyalur            67000 19-NOV-87 finance         assistant       s9
a10        mowni      m          madhavan   f   dharmapuri          75000 06-MAR-81 hr              manager         s10
```

9. Project only Name and DOB of dependent table.

```
SQL> select name,dob from dependent;

NAME                 DOB
-------------------- ---------
ezhil                01-DEC-89
pavan                02-JAN-92
sugan                18-JUN-52
prabha               27-APR-58
tamil                06-DEC-16
hari                 02-FEB-40
govind               25-MAR-93
samji                12-OCT-18
dev                  13-AUG-17
lassika              21-APR-57

10 rows selected.
```

10. Select the records of employee whose name start with “S”

```
SQL> select * from employee where fname like 's%';

SSN        FNAME      MINIT      LNAME      SEX ADDRESS            SALARY DOB          DEPARTMENT      DESIGNATION     SUPERVISOR
---------- ---------- ---------- ---------- --- --------------- ---------- --------- --------------- --------------- ----------
a1         sabari     j          eswar      m   cuddalore           50000 01-MAY-90 software            architect       a1
a2         sanjay     s          saravanan  m   salem               45000 12-JUN-89 ai          analyst         s2
a3         sanjika    s          senthil    f   krishnagiri         90000 24-MAY-89 hr          manager         s3
e02        sugan      r          rangan     m   vellore             49000 10-JAN-92 finance             manager         a02
e06        sara       l          davis      m   paris               65000 19-JUN-87 fianace             assistant       a06
```

11. Dependent names that are females and related to Employee no 1005

```
SQL> select name from dependent where relationship in('daughter','parent','mother')and employeessn='d4';

NAME
--------------------
prabha
```

12. Find the details of the projects that have been operating in the location ‘Chennai’

```
SQL> select * from project where location ='chennai';

PROJECT_NUM NAME                 LOCATION        CONTROLLING_DEPARTME     BUDGET
----------- -------------------- --------------- -------------------- ----------
          1 rocket               chennai         101                     200000
```

13. Display the Department details which has a manager.

```
SQL> select * from department where managerssn is not null;

no rows selected
```

14 . Select the employees who were born after 2010

```
SQL> select * from employee where dob>'01-jan-2010';

no rows selected

SQL> |
```

15.Display the employee names who salary is in the range 12000 to 30000

```
SQL> select  fname,minit,lname from employee where salary between 12000 and
30000;

no rows selected

SQL> |
```

16 . Find the employees who work for the project numbers 1, 3, and 45.

```
SQL> select * from project where project_num in(1,3,45);

PROJECT_NUM NAME                 LOCATION
   CONTROLLING_DEPARTME     BUDGET
----------- -------------------- ---------------------------------------
------- -------------------- ----------
          1 rocket               chennai
   101               200000
          3 lr023                kochi
   103               340000

SQL> |
```

17. Maximum salary withdrawn by an employee.

```
SQL> select max(salary) as max_salary from employee;

MAX_SALARY
----------
    300000
```

18. List the number of dependents for an employee.

```
SQL> select employeessn,count(*) as dependent_count from dependent group by
employeessn;

EMPLOYEESS DEPENDENT_COUNT
--------- ---------------
d1                      1
d10                     1
d2                      1
d3                      1
d4                      1
d5                      1
d6                      1
d7                      1
d8                      1
d9                      1

10 rows selected.
```

19.Print the total salary amount sanctioned in department no 4.

```
SQL> select sum(salary) as total from employee where ssn='a4';

     TOTAL
----------
     67000
```